

Time Step variations to save time, what effect does it have on the accuracy of computations?

The following data is related to an experiment (included in Models/experiments.py)
Running Glasberg on a basic signal with different values of timeStep that has
 $timeStep = 0,001$
as default value.

Successfully changed parameter timeStep to 0.001
File written successfully
Successfully changed parameter timeStep to 0.0015
File written successfully
Timer started for timeStep = 0.0015
Time for glasberg computation with timeStep = 0.0015 : 5.7108s
Successfully changed parameter timeStep to 0.0018
File written successfully
Timer started for timeStep = 0.0018
Time for glasberg computation with timeStep = 0.0018 : 4.5782s
Successfully changed parameter timeStep to 0.002
File written successfully
Timer started for timeStep = 0.002
Time for glasberg computation with timeStep = 0.002 : 4.3314s
Successfully changed parameter timeStep to 0.003
File written successfully
Timer started for timeStep = 0.003
Time for glasberg computation with timeStep = 0.003 : 3.6613s
Successfully changed parameter timeStep to 0.005
File written successfully
Timer started for timeStep = 0.005
Time for glasberg computation with timeStep = 0.005 : 3.1504s
Successfully changed parameter timeStep to 0.01
File written successfully
Timer started for timeStep = 0.01
Time for glasberg computation with timeStep = 0.01 : 2.4560s

